THE PROJECT EXPLAINED

Missing Data & Statistical Inference

A complete walkthrough from scratch: the methods, the bootstrap inside the Monte Carlo,
and how to explain every figure

Study Guide
M.Sc. Statistics
November 2026

How to Use This Guide

This guide explains the whole project in simple language, in the order the work was done. Every idea is explained the first time it appears, usually with a small example. By the end you should be able to explain any step, table or figure, and answer questions about it.

Chapter	What it gives you
1 The Project in One Page	30-second and 2-minute versions, the design diagram
2 Missing Data From Scratch	a ten-student example; MCAR, MAR, MNAR
3 The Statistical Ideas	bias, SE, intervals, coverage, bootstrap vs Monte Carlo, Rubin, PMM, EM, why bias limits coverage
4 What We Did, Step by Step	the notebook from Step 1a to Step 9g
5 Every Figure Explained	what each figure shows, how to read it, what to say
6 The Results in Plain Words	MCAR, MAR, MNAR as stories, and the table that explains it all
7 How to Present	a five-minute talk, the supervisor's points answered, do's and don'ts
8 Questions and Answers	about 30 likely questions
Appendices	glossary, formulas, notebook map, repository

The coloured boxes:

In simple words	a plain-language restatement
Example	a small worked example or analogy
Say it like this	a sentence you can say out loud
Watch out	a common mistake
Key idea	an idea worth remembering

- **One hour before a talk:** Chapters 1, 6, 7, 8.
- **To understand it:** Chapters 1–8 in order; read Chapter 4 next to the notebook.
- **To explain one figure:** Chapter 5 and Table 5.1.

Key idea. Three ideas carry the project: (1) the mechanism decides which methods can work; (2) the bootstrap makes one interval from one dataset, Monte Carlo tells us how often such intervals are right; (3) the bootstrap repairs over-confidence, not bias.

Contents

1 The Project in One Page **4**

1.1 The same thing in two minutes . . . 4
1.2 The roadmap . . . 5
1.3 Facts worth remembering . . . 6

2 Missing Data From Scratch **7**

2.1 What is missing data? . . . 7
2.2 Why does it matter? . . . 7
2.3 A tiny example you can check by hand . . . 7
2.4 The three kinds of missingness . . . 9

3 The Statistical Ideas, One at a Time **10**

3.1 Regression and the slope β_1 . . . 10
3.2 The true value versus the estimate . . . 10
3.3 Bias, spread and RMSE . . . 10
3.4 Standard error: how uncertain is one estimate? . . . 10
3.5 Confidence intervals and coverage . . . 11
3.6 The bootstrap: one dataset → one interval . . . 11
3.7 Monte Carlo: the true model → coverage . . . 12
3.8 Single imputation versus multiple imputation . . . 13
3.9 Rubin's rules . . . 13
3.10 Predictive mean matching (PMM) . . . 13
3.11 The EM algorithm . . . 13
3.12 The deletion curve (expit) . . . 14
3.13 The helper variable Z . . . 14
3.14 Why bias limits coverage . . . 14

4 What We Did, Step by Step **15**

4.1 Setup (Steps 1a–1b) . . . 15
4.2 Creating the complete data (Steps 2a–2e) . . . 15
4.3 The regression helper and the baseline (Steps 3a–3c) . . . 16
4.4 Deleting values (Steps 4a–4j) . . . 16
4.5 The eight methods on one dataset (Section 5) . . . 16
4.6 Building intervals on one dataset (Section 6) . . . 17
4.7 The Monte Carlo study (Section 7) . . . 18
4.8 Results and wrap-up (Sections 8–9) . . . 19
4.9 The safety checks built into the work . . . 19

5 Every Figure Explained **20**

5.1 The data (notebook Step 2e) . . . 20
5.2 The deletion rules (report Figure 1) . . . 21
5.3 How much is missing (Step 4i) . . . 21
5.4 What each method did to the data (Steps 5.10d–f; report Figure 2) . . . 22
5.5 One bootstrap (Step 6d; report Figure 3) . . . 23

5.6 Rubin’s rules in a picture (from Step 5.8f) . . . 24
5.7 EM convergence . . . 25
5.8 The Monte Carlo results, one mechanism (Steps 8g, 8j, 8m; report Figures 4–6) . . . 26
5.9 Individual intervals: classical vs bootstrap (Step 8n; report Figure 7) . . . 29
5.10 Before and after the bootstrap (Step 8o; report Figure 8) . . . 30
5.11 What the bootstrap changes, method by method (explainer only) . . . 31
5.12 Why bias limits coverage (explainer only; report Table 16) . . . 31
5.13 Where each figure is explained . . . 32

6 The Results in Plain Words 33

6.1 MCAR: the gaps are pure luck . . . 33
6.2 MAR: the gaps depend on something we observe . . . 34
6.3 MNAR: the gaps depend on the missing value itself . . . 34
6.4 The one table that explains it all . . . 35
6.5 What holds in every mechanism . . . 35
6.6 Why the one-dataset walkthrough is not evidence . . . 35

7 How to Present This Project 36

7.1 A five-minute talk . . . 36
7.2 The summary table to quote . . . 36
7.3 Explaining the design to your supervisor . . . 36
7.4 Do’s and don’ts . . . 37

8 Likely Questions and Model Answers 38

About the design . . . 38
About intervals and coverage . . . 38
About the methods . . . 39
About the conclusions . . . 40

Appendix A: Glossary 41

Appendix B: Formula Sheet 42

Appendix C: Notebook Map 42

Appendix D: The Repository 42

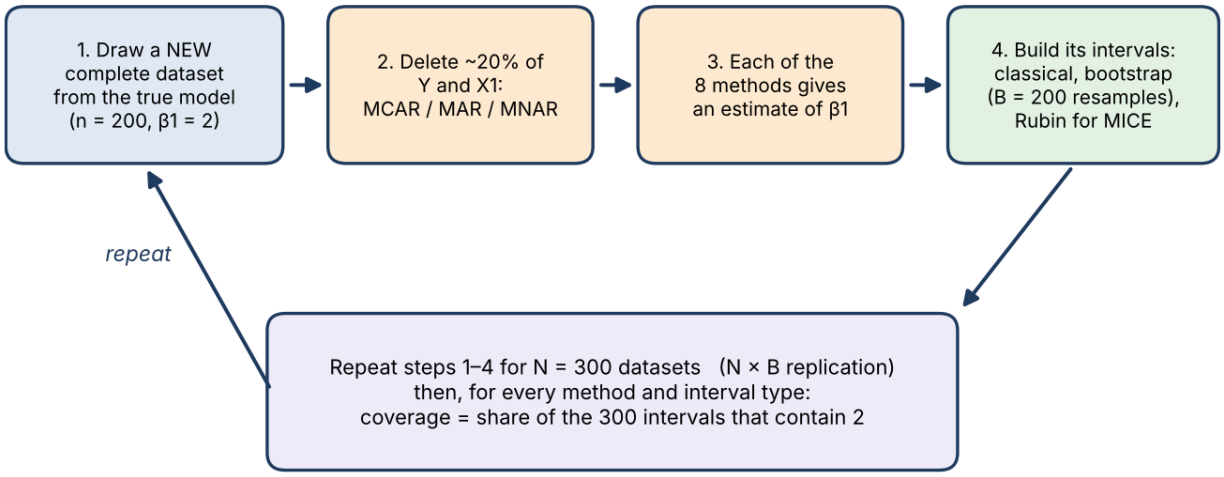
1 The Project in One Page

The whole project in 30 seconds. We wanted to know what happens to a regression when some data are missing: does the way we handle the gaps change the answer, and can we still trust the confidence interval? We made up data from a model where the true slope is known (β1 = 2), deleted values in three ways — by pure chance, depending on another variable we can still see, and depending on the missing value itself — and tried eight methods. For every method we built the confidence intervals an analyst would use: the classical one, two bootstrap ones and, for multiple imputation, Rubin's. Then we ran a Monte Carlo study: 300 new datasets, with 200 bootstrap resamples inside each, and counted how often each interval contained the truth. We found that mean imputation is always bad; that classical intervals after single imputation are too narrow; that the bootstrap fixes that, but cannot fix a biased estimate; that MICE and EM work when the missingness is random or depends on something we observe; and that when it depends on the missing value itself, nothing works reliably.

1.1 The same thing in two minutes

1. **The problem.** Real data often have empty cells. Quick fixes — delete the incomplete rows, fill gaps with the average — are easy, but they can bias the results and make us *too sure* of them.
2. **The experiment.** With real data we never learn the true answer, so we *make up* data from a known formula, $Y = 5 + 2X_1 - 1.5X_2 + X_3 + \text{error}$, with 200 rows. The number we care about is $\beta_1 = 2$. About 20% of the values of Y and of X_1 are deleted.
3. **Three ways of deleting.** MCAR: pure chance. MAR: the chance depends on a helper variable Z that stays visible. MNAR: the chance depends on the value that disappears. Methods are compared *inside* each of the three.
4. **Eight methods.** Delete incomplete rows; fill with the mean; copy from a similar row (hot-deck); fill with a regression prediction; prediction plus noise (stochastic regression); MICE in two versions (10 completed datasets combined with Rubin's rules); EM (maximum likelihood).
5. **Four kinds of 95% interval.** Classical (estimate ± 1.96 × formula SE), bootstrap standard (± 1.96 × bootstrap SE), bootstrap percentile (middle 95% of the bootstrap estimates), and Rubin's interval for MICE.
6. **Bootstrap inside Monte Carlo.** The bootstrap turns *one* dataset into *one* interval. Monte Carlo draws *many* datasets from the true model, so we can count how often the intervals contain the truth (the **coverage**). We do both: 300 datasets × 200 bootstrap resamples (N × B replication).
7. **What we found.** Mean imputation is worst everywhere. Classical intervals after single imputation are too narrow. The bootstrap repairs the width but not a biased estimate — the remaining shortfall is predicted almost exactly by the bias. MICE and EM are good under MCAR and MAR. Under MNAR nothing reaches 95%.

1.2 The roadmap



Before the study: steps 2–4 are shown on ONE dataset (seed 42) only to SEE what each method does.

Figure 1.1: The study. One repetition = steps 1–4; the whole study repeats them 300 times.

Part	What happens	Notebook
Setup and data	Settings; one complete dataset; the regression helper; three deletion rules.	Sections 1–4
Walkthrough	Every method applied to ONE dataset, to see what it does (no conclusions).	Section 5
Intervals	Classical, bootstrap and Rubin intervals on one dataset; why one interval cannot show coverage.	Section 6
Monte Carlo	300 new datasets × 3 mechanisms × 8 methods, with 200 bootstrap resamples inside each.	Section 7
Results	Bias, accuracy and coverage of every interval, one mechanism at a time.	Section 8
Wrap-up	Master tables, best method per mechanism, findings, limitations, conclusion.	Section 9

Table 1.1: How the notebook is organised.

1.3 Facts worth remembering

Fact	Value
True model	Y = 5 + 2·X1 − 1.5·X2 + 1·X3 + error; Z = Y + noise
Number we care about	β1 = 2
Rows per dataset	200
Missing	about 20% of Y and of X1 (about a third of rows have a gap)
Mechanisms	MCAR, MAR (driven by Z), MNAR (driven by the value itself)
Methods	8: complete case, mean, hot-deck, regression, stochastic regression, MICE normal, MICE PMM, EM
Intervals	classical · bootstrap standard · bootstrap percentile · Rubin (MICE)
Monte Carlo	300 datasets, seed 2024
Bootstrap	200 resamples inside every dataset (N × B)
MICE	10 imputations
Benchmark	with nothing missing, classical coverage 94.3%, bootstrap 93.7% (both consistent with 95%)
Random error of a coverage	about ±2.5 points (95% band)

Table 1.2: Cheat card.

2 Missing Data From Scratch

2.1 What is missing data?

A dataset is a table: each row is a person (or machine, or day) and each column something measured. **Missing data** means that some cells are empty. In Python an empty cell is shown as **NaN** (“not a number”).

Person	Age	Income (₹ thousand)
1	25	30
2	31	45
3	40	*missing*
4	52	70
5	28	38

Table 2.1: One empty cell — the whole problem in miniature.

2.2 Why does it matter?

- **Less information**: fewer complete rows means less precise answers.
- **What we do about the gap can change the answer**, e.g. filling with an average quietly weakens relationships.
- **How much it changes depends on *why* the value is missing** — the most important idea in the project.

2.3 A tiny example you can check by hand

Ten students studied 1 to 10 hours and took an exam. **How many extra marks does one more hour give?** That is the slope of a line through the points. Suppose the scores of students 3, 6 and 9 get lost.

Student	Hours	Real score	Score we see	After mean imputation	After regression imputation
1	1	57	57	57	57
2	2	54	54	54	54
3	3	64	lost	70.7	61.9
4	4	61	61	61	61
5	5	74	74	74	74
6	6	71	lost	70.7	73.5
7	7	83	83	83	83
8	8	80	80	80	80
9	9	89	lost	70.7	85.0
10	10	86	86	86	86

Table 2.2: The ten students.

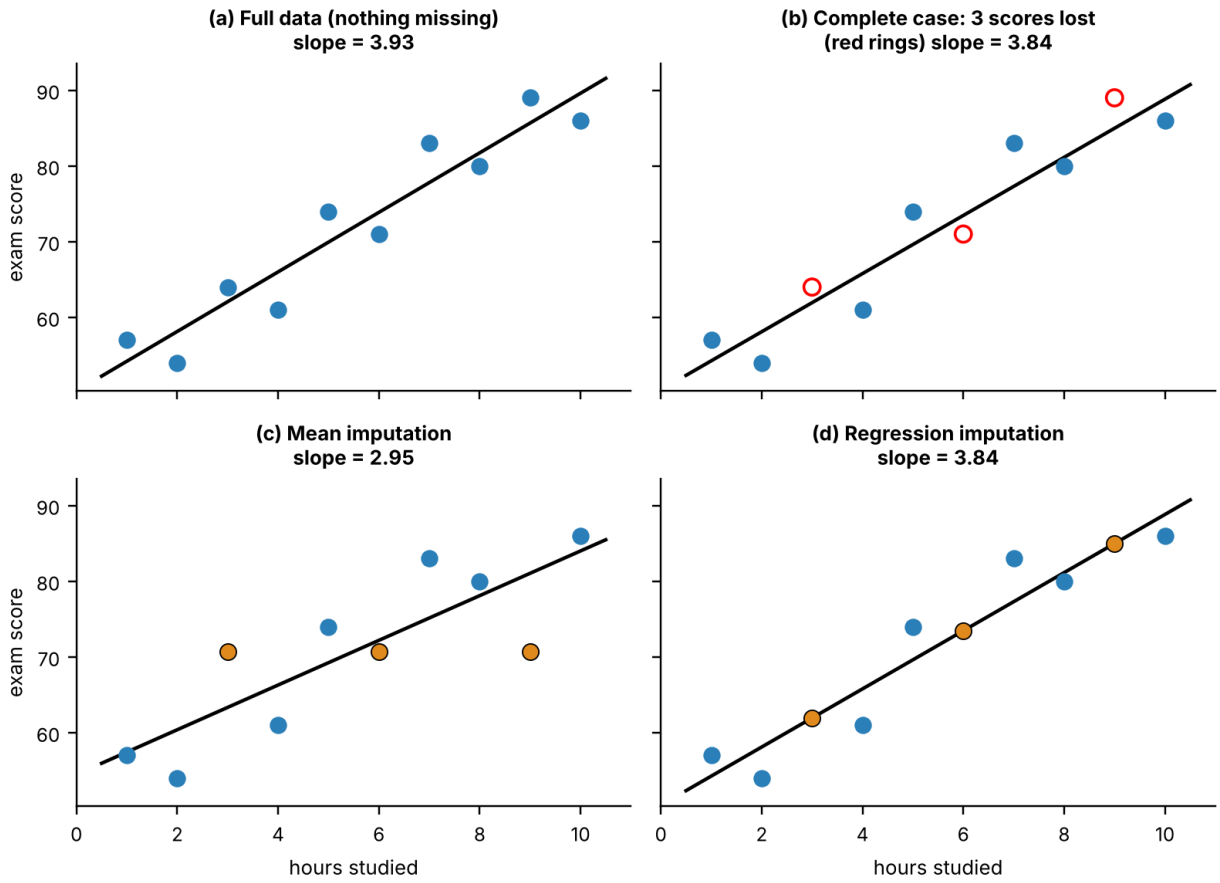


Figure 2.1: (a) truth; (b) delete the three students; (c) fill with the mean; (d) fill with the regression prediction.

- **(a) Full data:** slope **3.93** — the truth to recover.
- **(b) Complete case:** slope **3.84**. Lost by luck, so the remaining students tell the right story; we just have less data.
- **(c) Mean imputation:** all three gaps get 70.71; students with 3 and 9 hours get the same score. The slope drops to **2.95**.
- **(d) Regression imputation:** the filled points sit exactly on the line (slope 3.84, as in (b)). They add no information but make the data *look* like ten real points — which will make us too confident.

Missing for a reason. Now suppose the three students who did worse than their hours predicted (students 2, 4 and 10) were embarrassed and did not report. The average of the reported scores is 74.0, but the true average is 71.9. Nothing in the reported scores reveals that — the idea behind MNAR.

Watch out. This example is made up to show the ideas; it is not evidence. The evidence is the simulation.

2.4 The three kinds of missingness

Name	Chance of missing depends on…	Everyday example	In our simulation
MCAR Missing Completely At Random	nothing	A sensor fails at random moments.	every value has a 20% chance
MAR Missing At Random	something we *do* observe	Older people skip an income question; age is recorded for all.	the chance depends on Z (always observed)
MNAR Missing Not At Random	the missing value itself	Very high earners refuse to report income.	large Y and large X1 vanish more often

Table 2.3: MCAR, MAR and MNAR.

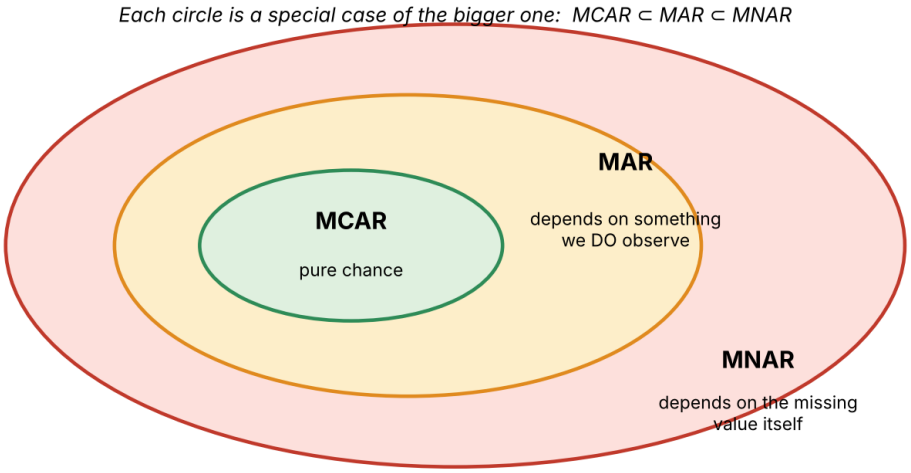


Figure 2.2: The three mechanisms are nested.

How to read Figure 2.2. MCAR (green) is the most special case; MAR (orange) contains it (dependence on observed variables is allowed, and MCAR is the case where it is zero); MNAR (red) is the most general. MCAR ⊂ MAR ⊂ MNAR.

Key idea. MAR versus MNAR **cannot be tested from the observed data**: the difference lies in the values we do not see. We know which one we are in only because we wrote the deletion rule.

3 The Statistical Ideas, One at a Time

3.1 Regression and the slope β₁

A regression finds the best-fitting line (or plane). In $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \text{error}$, each β says how much Y changes when its X rises by one, the others held fixed. We chose $\beta_1 = 2$, so one more unit of X_1 adds 2 to Y on average. We focus on β_1 because X_1 itself loses values.

3.2 The true value versus the estimate

A sample is a random draw, so even with nothing missing the estimate is not exactly the truth. Our walkthrough dataset gives **2.037**. That is why one dataset can never tell us whether a method works — and why we need many datasets.

3.3 Bias, spread and RMSE

- **Bias**: is the average estimate on target? (average of estimate − 2)
- **Spread (SD)**: how much do estimates wobble from dataset to dataset?
- **RMSE**: overall error, $\text{RMSE}^2 \approx \text{bias}^2 + \text{spread}^2$.

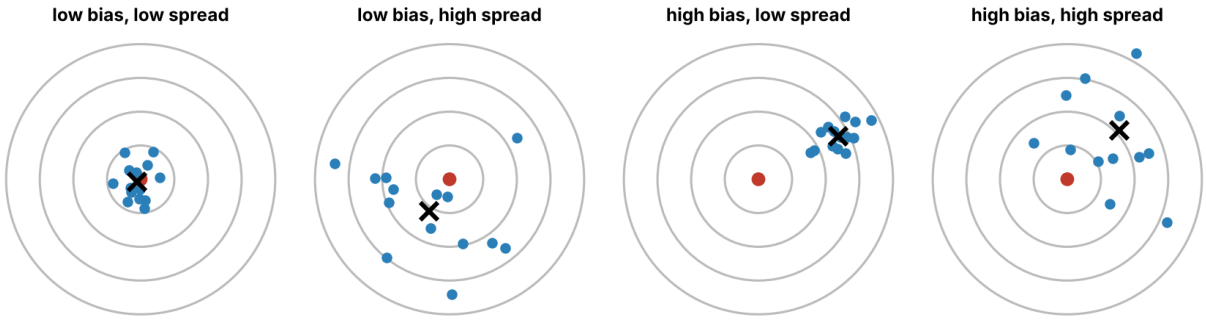


Figure 3.1: Bias and spread as a dartboard.

The red dot is the truth, the blue dots are estimates from different datasets, the × is their average. Bias is how far the × is from the red dot; spread is how scattered the dots are. High bias with low spread (third board) is the most dangerous: consistent, therefore convincing, but wrong.

Check the formula. Complete case under MAR: bias -0.103, spread 0.100. $\sqrt{(-0.103^2 + 0.100^2)} = 0.144$; the RMSE in the results is 0.144.

3.4 Standard error: how uncertain is one estimate?

The **standard error (SE)** is the typical size of the estimate's wobble. In a simulation we can see the true wobble (the SD of the estimates over many datasets) and compare it with the SE each method *reports*. If the reported SE is smaller, the method is **over-confident**.

Classical SE comes from the regression formula. After single imputation it is *naive*: it treats made-up values as real. For hot-deck, regression and stochastic regression it is only

68–79% of the true spread.

3.5 Confidence intervals and coverage

A **95% confidence interval** is a range that is supposed to catch the true value 95 times out of 100 — not in one dataset, but over many repetitions.

> **The net analogy.** Think of fishing nets. Throw 100 nets (one per dataset). A good 95% net catches the fish (the true value) in about 95 throws. A net that catches it 60 times is too small or thrown in the wrong place, and its "95%" label is false.

Coverage = the share of the 300 intervals that contain 2. With nothing missing, the classical interval covers 94.3%, as a correct simulation should. An interval can fail in two ways: it can be **centred in the wrong place** (bias) or be **too narrow** (over-confident SE). Coverage catches both.

3.6 The bootstrap: one dataset → one interval

The bootstrap estimates the SE from a single dataset by **treating the sample as if it were the population**: (1) draw rows *with replacement* until you have a new dataset of the same size; (2) redo the analysis — here, **the whole method, including the filling-in of gaps**; (3) repeat B times (we use 200); (4) the SD of the B estimates is the bootstrap SE.

Resample	Numbers drawn	Average
1	6 4 4 7 4	5.0
2	6 6 5 4 4	5.0
3	7 5 9 5 7	6.6
4	4 9 9 4 4	6.0
5	5 7 6 5 5	5.6
6	5 9 5 4 9	6.4

Table 3.1: Six bootstrap resamples of the numbers 4, 7, 5, 9, 6.

Because we draw with replacement, a number can appear twice and another not at all. The averages vary; their SD over 5000 resamples is 0.78 — the bootstrap SE of the average (the textbook formula gives 0.86; the gap is normal for 5 numbers).

Four kinds of bootstrap interval

Interval	How it is built	Used here?
Standard	estimate ± 1.96 × bootstrap SE	yes
Percentile	the 2.5% and 97.5% points of the B bootstrap estimates	yes
Bootstrap-t	like the standard one, but uses a t-like table built by bootstrapping inside each bootstrap	no (needs N × B × B′ fits)
BCa	percentile interval corrected for bias and skewness	no

Table 3.2: Bootstrap confidence intervals. Only the standard and bootstrap-t intervals use the bootstrap SE.

3.7 Monte Carlo: the true model → coverage

A **Monte Carlo simulation** draws many new datasets *from the true population* (here, the known model). Because the truth is known, we can count how often each interval contains it. **Without Monte Carlo, the coverage probability of an interval cannot be found.**

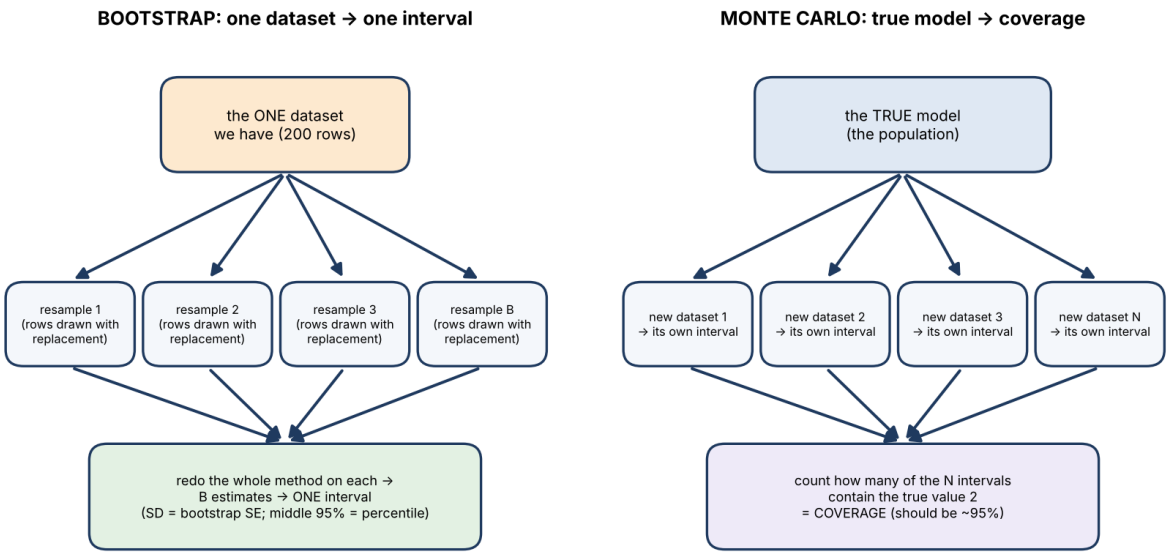


Figure 3.2: Bootstrap versus Monte Carlo.

	Bootstrap	Monte Carlo
Starts from	the one sample we have	the true model (population)
Creates	resamples of the same rows	brand-new datasets
Knows the truth?	no	yes
Gives	one SE and one interval	the coverage of a whole type of interval
In this project	200 resamples inside each dataset	300 datasets

Table 3.3: The two are different things — and both are needed.

Key idea. Bootstrap inside Monte Carlo (N × B). For each of the N = 300 Monte Carlo datasets, the bootstrap is run B = 200 times to make that dataset's interval. Then the N intervals are checked against the truth. One bootstrap on its own gives one interval and tells us nothing about coverage.

Monte Carlo error. With only 300 repetitions, a coverage figure has its own random error: about $\sqrt{(0.95 \times 0.05 / 300)} \approx 1.3$ points (standard error), so ±2.5 points for a 95% band. Differences smaller than that should not be over-read. A **random seed** fixes the random numbers so anyone gets exactly the same results; each repetition has its own seed, so results are identical whether run on one core or many.

3.8 Single imputation versus multiple imputation

Single imputation fills each gap once and analyses the data as if the filled values were real. **Multiple imputation** fills each gap several times, analyses each completed dataset, and combines the results.

The homework analogy. Copying a friend's homework for the questions you skipped, then telling the teacher you answered everything yourself: the teacher would trust your score more than they should. Single imputation does this to the standard error.

Rough rule. The SE shrinks like 1/√(number of real observations). If 30% of rows contain made-up values, the formula behaves as if it had 100 real rows when there are about 70, so the naive SE is about √0.7 ≈ 84% of the truth; with 39% of rows affected, about 78%. In our study 30–39% of rows have a gap, and the measured classical ratios were 68–79%.

3.9 Rubin's rules

- **Estimate** = average of the m estimates.
- **Within variance W** = average of the m ordinary variances (SE^2).
- **Between variance B** = variance of the m estimates (how much they disagree).
- **Total** $T = W + (1 + 1/m)\cdot B$; pooled SE = $\sqrt{T}$. The interval is estimate ± 1.96 × $\sqrt{T}$ (the **Rubin interval**).

Rubin by hand (m = 3). Three completed datasets give 1.80, 2.00, 1.90, each with SE 0.10. Average 1.90; W = 0.0100; B = 0.0100; T = 0.0100 + (4/3)·0.0100 = 0.0233; pooled SE = **0.153**. Ignoring the disagreement would claim 0.100.

3.10 Predictive mean matching (PMM)

PMM fills a gap with a *real observed value*: predict the missing value, find the 5 observed rows whose predictions are closest, pick one at random and copy its **actual** value. Filled values look realistic, but PMM can only borrow values that exist — a weakness when a row's prediction is extreme and has few close donors.

3.11 The EM algorithm

EM finds maximum-likelihood estimates from incomplete data by repeating two steps until nothing changes: the **E-step** fills each gap with its expected value given the observed values (keeping track of the uncertainty left), and the **M-step** re-estimates the means and covariances. The regression coefficients then follow: $\beta = \Sigma_{xx}^{-1}\Sigma_{xy}$. EM has no simple SE formula, so it gets only bootstrap intervals.

In simple words. Like redrawing a map from a rough sketch: guess the missing parts, redraw, improve the guesses with the new map, repeat until the map stops changing.

3.12 The deletion curve (expit)

expit(z) = $1/(1 + e^{-z})$ turns any number into a probability. For MAR and MNAR the chance of deletion is expit($-2.23 + 2 \times d$), where d is the standardised driver (Z for MAR; the value itself for MNAR). The -2.23 keeps the average near 20%.

Driver d	-2	-1	0	1	2	3
Chance of deletion	0%	1%	10%	44%	85%	98%

Table 3.4: The deletion chance for different drivers.

3.13 The helper variable Z

$Z = Y +$ noise is observed for everyone but is not in the regression. Like a cheap bathroom-scale reading next to an exact weight, it tells us a lot about a missing Y. Under MAR, rows with large Z lose values; methods that use Z (imputation methods, EM) can correct for this, complete case cannot.

3.14 Why bias limits coverage

If an estimate is normal with bias b and spread SD, and the interval uses an SE equal to r × SD, the interval contains the truth with probability

$$\Phi(1.96 \cdot r - b/SD) - \Phi(-1.96 \cdot r - b/SD),$$

where Φ is the normal distribution function. With an honest SE (r = 1) and no bias this is 95%. As |bias|/SD grows, coverage falls — no matter how honest the SE is.

Bias ÷ SD	0	0.5	1.0	1.5	2.0	3.0
Coverage with honest SE	95%	92%	83%	68%	48%	15%
Coverage with SE = 0.75 × SD	86%	81%	67%	49%	30%	6%

Table 3.5: How bias and an over-confident SE reduce coverage.

Key idea. The bootstrap can make the SE honest (r ≈ 1). It cannot make the bias zero. That single sentence explains most of the results.

4 What We Did, Step by Step

This chapter follows the notebook from top to bottom. Every code block in the notebook starts with a label such as **Step 2a** or **Step 7d**. For each step you get what it does, why, and what you see. Read it with the notebook open and you can explain any block.

4.1 Setup (Steps 1a–1b)

1a loads the tools: numpy and pandas (numbers and tables), matplotlib (plots), statsmodels (only to double-check our regression), scipy (the expit curve), scikit-learn (the nearest-neighbour search for hot-deck). **1b** holds every setting in one place:

Setting	Value	Meaning
SEED	42	seed for the one-dataset walkthrough (Sections 2–6)
N	200	rows per dataset
TRUE_BETA	5, 2, −1.5, 1	true intercept and slopes
M_IMPUTATIONS	10	completed datasets in MICE
N_REPS	300	Monte Carlo repetitions (new datasets)
B_BOOT	200	bootstrap resamples inside every dataset
MC_SEED	2024	seed for the Monte Carlo study
N_JOBS	−1	CPU cores to use (−1 = all; 1 = no parallel computing)
USE_SAVED_RESULTS	True	reuse saved results if they exist with the same settings
RESULTS_DIR	results/	where results are saved

Table 4.1: The settings block (Step 1b). For a quick test: N_REPS = 20, B_BOOT = 50.

4.2 Creating the complete data (Steps 2a–2e)

2a defines *make_data*: three standard-normal predictors, $Y = 5 + 2{\cdot}X1 - 1.5{\cdot}X2 + X3 +$ noise, and $Z = Y +$ noise. **2b** creates the walkthrough dataset (seed 42). **2c** prints summary statistics, **2d** the correlations, **2e** draws the data (Figure 5.1).

	X1	X2	X3	Z	Y
X1	1.00	0.05	-0.08	0.68	0.70
X2	0.05	1.00	0.07	-0.43	-0.48
X3	-0.08	0.07	1.00	0.17	0.23
Z	0.68	-0.43	0.17	1.00	0.94
Y	0.70	-0.48	0.23	0.94	1.00

Table 4.2: Correlations (Step 2d). Z and Y are very strongly related.

4.3 The regression helper and the baseline (Steps 3a–3c)

3a defines *ols*, the regression every method ends with: $\beta = (A^TA)^{-1}A^Ty$ and its standard errors. **3b** checks it against statsmodels (prints True). **3c** fits it to the complete data — the best any method could match:

Parameter	True	Estimate	SE
β0	5.000	4.9629	0.0726
β1	2.000	2.0368	0.0722
β2	-1.500	-1.5076	0.0749
β3	1.000	0.9483	0.0776

Table 4.3: The baseline (Step 3c).

4.4 Deleting values (Steps 4a–4j)

4a–4c define the three deletion rules, **4d** picks one by name, **4e–4g** make the three walkthrough datasets, **4h** counts the gaps, **4i** draws them (Figure 5.3), **4j** checks the rules on 20,000 rows.

	MCAR	MAR	MNAR
X1 missing	45	39	41
Y missing	43	40	41
Rows with a gap	78	61	65
Complete rows	122	139	135

Table 4.4: Gaps in the walkthrough datasets (Step 4h).

Mechanism	Y missing %	X1 missing %	Rows with gap %	Z gap	Y gap
MCAR	20.11	20.02	36.10	-0.01	-0.03
MAR	19.87	20.16	30.58	3.96	3.53
MNAR	19.82	20.08	32.28	3.74	3.72

Table 4.5: Check of the rules on 20,000 rows (Step 4j). 'Gap' = average among rows that lost Y minus the rest: about zero under MCAR, clearly positive under MAR and MNAR.

4.5 The eight methods on one dataset (Section 5)

Each method has its own sub-section: **a** defines it, **b, c, d** apply it to the MCAR, MAR and MNAR datasets and show what it filled in. **This section is an illustration only** — one dataset is one random draw. The helpers **5.1a** (stores the estimates) and **5.1b** (shows true values next to filled values) come first.

Steps	Method	What the blocks show
5.2a–d	Complete case	how many rows are kept (122, 139, 135 of 200)
5.3a–d	Mean imputation	the means used; every gap gets the same number
5.4a–d	Hot-deck	values copied from the nearest row (scikit-learn, one neighbour)
5.5a–d	The imputation engine	regress_column, draw_new_coefficients (MICE), pmm_donors (MICE), chained_impute (cycles 5 times)
5.6a–d	Regression imputation	filled values lie on the regression line
5.7a–d	Stochastic regression	prediction + random residual
5.8a–j	MICE normal and PMM	impute 10 times (a), fit each (b), pool with Rubin (c); the 10 estimates and the within/between variances
5.9a–f	EM	the algorithm (a), coefficients from means and covariances (b), check that EM = OLS with nothing missing (c), EM means vs observed means

Table 4.6: Section 5 at a glance.

Real numbers (Step 5.8f, MICE normal, MAR): the 10 estimates range from 1.930 to 2.102, average 2.036; W = 0.00583, B = 0.00349, total = W + 1.1·B = 0.00967, pooled SE = 0.098.

5.10a puts side by side what every method filled into the same gaps; **5.10b** lists every method's estimate on this dataset; **5.10c–f** draw the data after each method (Figure 5.4).

Method	MCAR	MAR	MNAR
Complete Case	2.098	1.844	2.015
Mean Imputation	1.622	1.346	1.263
Hot-Deck	2.005	1.878	1.935
Regression Imputation	2.168	2.110	2.212
Stochastic Regression	2.060	2.085	2.120
MICE (normal)	2.082	2.036	2.107
MICE (PMM)	2.062	1.943	2.047
EM	2.076	2.022	2.108

Table 4.7: Estimates on the one walkthrough dataset (Step 5.10b). Complete-data estimate 2.037.

4.6 Building intervals on one dataset (Section 6)

6a defines *bootstrap_beta1*: resample the rows with replacement and run the **whole** method B times. **6b** defines the two interval recipes: *normal_interval* (estimate ± 1.96 × SE — used for classical, bootstrap-standard and Rubin) and *percentile_interval*. **6c** builds every interval of every method on the MAR dataset:

Method	Estimate	Classical	Bootstrap standard	Bootstrap percentile	Rubin
Complete Case	1.844	(1.639, 2.050)	(1.643, 2.045)	(1.660, 2.054)	–
Mean Imputation	1.346	(1.084, 1.607)	(1.054, 1.637)	(1.066, 1.596)	–
Hot-Deck	1.878	(1.713, 2.042)	(1.613, 2.143)	(1.604, 2.101)	–
Regression Imputation	2.110	(1.977, 2.244)	(1.927, 2.294)	(1.942, 2.257)	–
Stochastic Regression	2.085	(1.930, 2.240)	(1.913, 2.257)	(1.862, 2.211)	–
MICE (normal)	2.036	–	–	–	(1.843, 2.228)
MICE (PMM)	1.943	–	–	–	(1.749, 2.137)
EM	2.022	–	(1.856, 2.188)	(1.856, 2.195)	–

Table 4.8: All intervals on the MAR walkthrough dataset (Step 6c, 200 resamples).

6d draws the bootstrap of stochastic regression (Figure 5.5). The section ends with the key point: **one interval cannot tell us whether a type of interval works** — each either contains 2 or not, by luck. That needs Monte Carlo.

4.7 The Monte Carlo study (Section 7)

7a collects the methods and shows which intervals each gets (classical for complete case and single imputation; bootstrap for those and EM; Rubin for MICE). **7b** defines *one_replication*: one new dataset from the true model; for each mechanism, delete values; run every method; record its estimate, classical SE, Rubin SE, and bootstrap SE and percentile limits from B resamples; also record the complete-data benchmark (with its own bootstrap). **7c** runs one replication so you can see the 27 records.

7d runs all 300 repetitions. Four practical features: (1) each repetition gets its own seed from *SeedSequence(2024).spawn(300)*, so results do not depend on how many cores are used; (2) repetitions run in parallel with *joblib* (N_JOBS = −1 uses all cores); (3) progress is saved every 10 repetitions to *monte_carlo_partial.csv*, so an interrupted run resumes where it stopped; (4) the final results are saved, and a later run with the same settings simply loads them. The full study took about 21 minutes on 2 cores.

Column	Meaning
Mechanism, Method, Rep	which setting, method and repetition
Beta1	the estimate
Beta1_full	the estimate from the same dataset with nothing missing
SE_classical	formula SE (complete case, single imputation, benchmark)
SE_rubin	Rubin’s-rules SE (MICE)
SE_boot	SD of the 200 bootstrap estimates
Pct_low, Pct_high	2.5% and 97.5% points of the bootstrap estimates

Table 4.9: What is saved for every record (results/monte_carlo_results.csv, 8100 rows).

7e turns every record into its intervals and marks whether each contains 2 and how wide it is. **7f** summarises the estimates (bias, bias vs full data, SD, RMSE). **7g** summarises the intervals: coverage, average width, and reported SE ÷ true spread.

4.8 Results and wrap-up (Sections 8–9)

8a–8d define two tables and three figures; **8e–8m** show them for MCAR, MAR and MNAR; **8n** draws individual intervals (classical vs bootstrap percentile); **8o** draws coverage heat maps without and with the bootstrap. **9a–9e** are master tables, a scoreboard and the best method per mechanism; **9f** saves the CSV files; **9g** writes findings automatically from the numbers, so text and tables cannot disagree. The interpretation, limitations and conclusion follow.

4.9 The safety checks built into the work

- Our regression matches statsmodels exactly (Step 3b).
- EM with nothing missing reproduces ordinary regression (Step 5.9c).
- The deletion rules are verified on 20,000 rows (Step 4j).
- With nothing missing, the classical interval covers 94.3% and the bootstrap percentile interval 93.7% — both consistent with 95%, so the simulation and the bootstrap work.
- Each repetition has its own seed; an interrupted run resumed and gave identical results (tested).
- Findings (Step 9g) and the report text are generated from the results; the report build stops if a claim no longer matches the data.

5 Every Figure Explained

Each figure gets a card: **what it shows**, **how to read it**, **what we see**, **why**, a line to **say**, and a likely question. Colours are the same everywhere: **grey** = complete case, **orange** = single imputation (mean, hot-deck, regression, stochastic), **blue** = MICE and EM, **light grey** = the benchmark with nothing missing. For interval types: **orange** = classical, **light blue** = bootstrap standard, **dark blue** = bootstrap percentile, **green** = Rubin. A red dashed line marks the truth, $\beta_1 = 2$.

5.1 The data (notebook Step 2e)

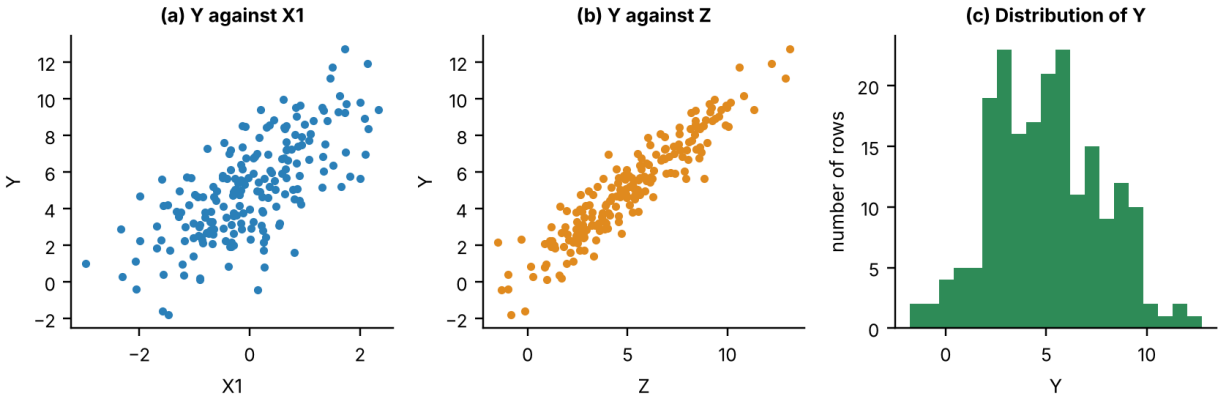


Figure 5.1: The complete walkthrough dataset.

What it shows	Three views of the 200-row dataset before anything is deleted.
How to read it	(a) each dot is a row, X1 across, Y up; (b) the same with Z across; (c) a histogram counting rows in each range of Y.
What we see	(a) Y rises with X1 with lots of scatter; (b) Y and Z almost on a line (correlation 0.94); (c) Y is roughly bell-shaped (mean 5.1, SD 2.7).
Why	Y depends on X1 but also on X2, X3 and noise, hence the scatter; Z is Y plus small noise.
Say it	“Before deleting anything we check the data look as designed: Y follows X1, Z is a close proxy for Y.”
If asked	**Is such a strong Z realistic?** Usually not — real helper variables are weaker. We state this as a limitation.

5.2 The deletion rules (report Figure 1)

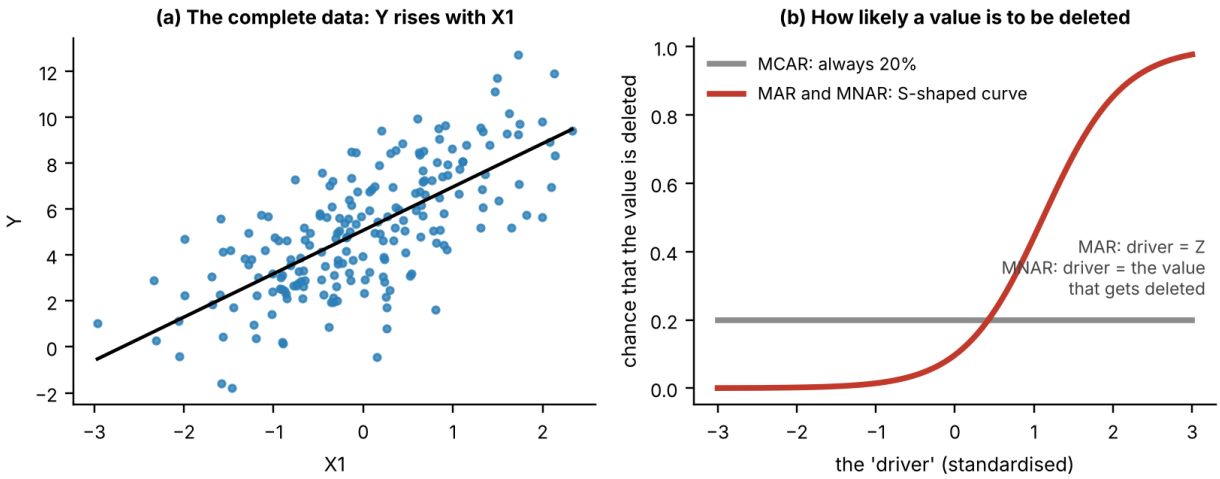


Figure 5.2: (a) the complete data with its fitted line; (b) chance of deletion against the driver.

How to read it	In (b) the horizontal axis is the driver (standardised: average 0, SD 1), the vertical axis the probability of deletion. Grey flat line = MCAR. Red S-curve = MAR and MNAR; they differ only in what the driver is (Z for MAR, the value itself for MNAR).
What we see	MCAR: 20% for everyone. MAR/MNAR: 10% at an average driver, 44% one SD above, 85% two SDs above, almost 0% far below.
Why	The −2.23 keeps the overall rate near 20%, so all mechanisms delete the same *amount* and differ only in *who* loses values.
Say it	“All three delete about 20%; what differs is who loses values: nobody in particular, rows with large Z, or rows with large values.”

5.3 How much is missing (Step 4i)

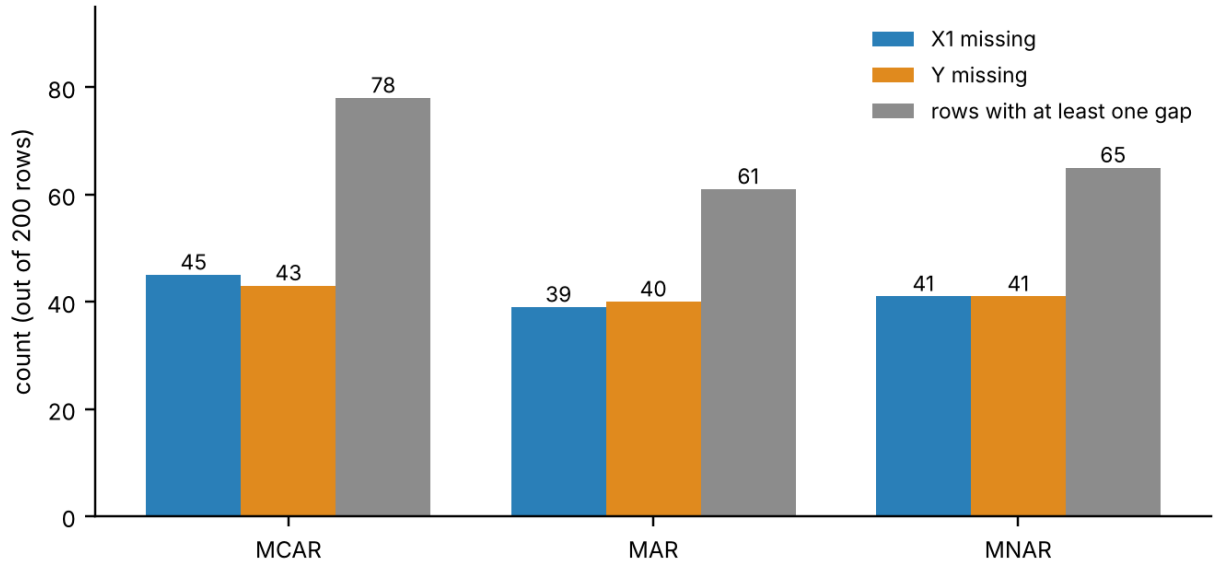


Figure 5.3: Missing values in the three walkthrough datasets.

How to read it	Three groups (mechanisms); bars = missing X1 (blue), missing Y (orange), rows with any gap (grey); numbers on top.
What we see	Rows with a gap: 78 (MCAR), 61 (MAR), 65 (MNAR) of 200.
Why	Each deletion is random, so counts are near 40 but not exactly; the grey bar counts a row once even if it lost both values.
Say it	“About a fifth of each variable, so roughly a third of the rows, have a gap — that is why the method matters.”

5.4 What each method did to the data (Steps 5.10d–f; report Figure 2)

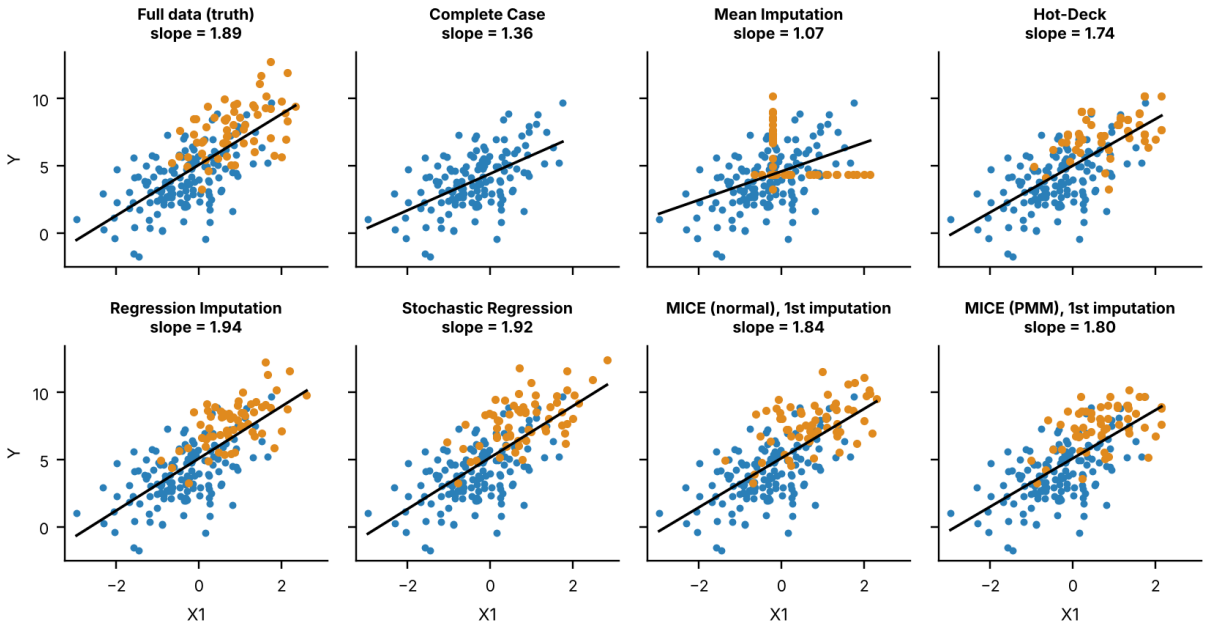


Figure 5.4: Y against X1 after each method handled the gaps (MAR walkthrough dataset).

How to read it	Each panel is the same dataset after one method. Blue = rows without a gap; orange = rows that had a gap (filled in, or — for complete case — removed). The black line and the slope in the title are for Y on X1 alone (full data: 1.89); compare panels with each other.
What we see	Complete case loses the top-right of the cloud (slope 1.36); mean imputation draws a cross through the average (1.07); regression imputation puts the orange points exactly on the line; stochastic regression and MICE scatter them like real data.
Why	Under MAR, rows with large Z (hence large Y) lose values, so deleting them flattens the line. Mean imputation ignores X1 and Z, so the filled points cannot follow the trend.
Say it	“One picture shows why each method behaves as it does: a missing corner, a cross, a line, or realistic scatter.”
If asked	**Why the top-right?** Under MAR the rows with large Z lose values, and large Z means large Y and, through the model, large X1.

5.5 One bootstrap (Step 6d; report Figure 3)

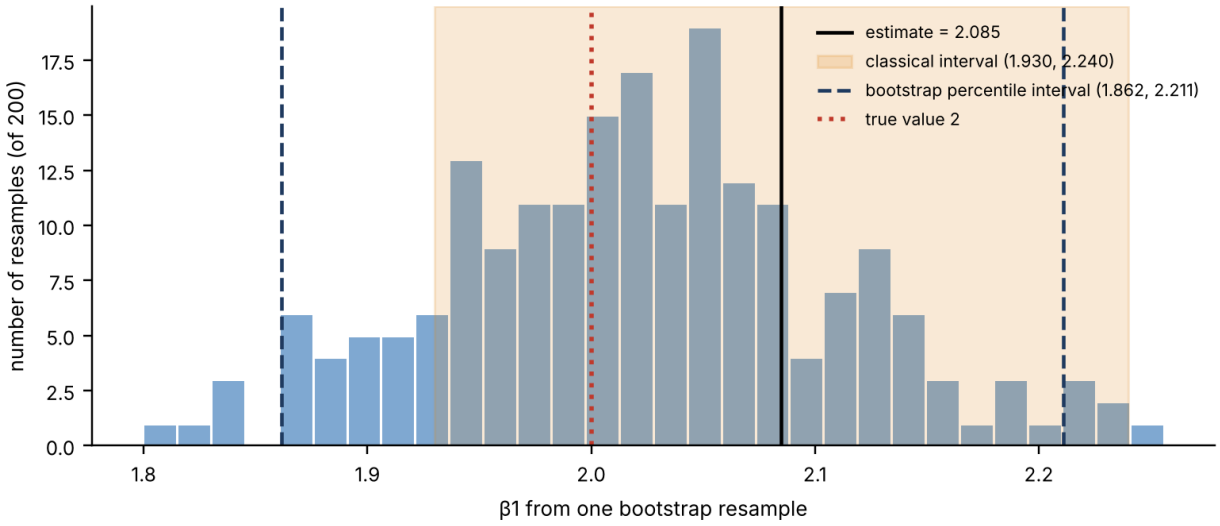


Figure 5.5: 200 bootstrap estimates of β1 for stochastic regression on the MAR walkthrough dataset.

What it shows	What one bootstrap produces: 200 estimates, each from a resample of the same 200 rows with the whole method re-run.
How to read it	The histogram counts resamples in each narrow range of β1. Black line = the estimate on the actual data. Orange band = the classical interval. Dashed dark-blue lines = the bootstrap percentile interval (2.5% and 97.5% points). Red dotted = truth.
What we see	Estimate 2.085; classical interval (1.930, 2.240); percentile interval (1.862, 2.211); bootstrap SE 0.088. Both intervals happen to contain 2 here.
Why	The bootstrap repeats the filling-in each time, so its spread includes the uncertainty created by imputation; the classical formula does not.
Say it	“The bootstrap: resample, redo everything, look at the spread. But this is one interval from one dataset — it can’t tell us how often such intervals are right.”
If asked	**So did the bootstrap ‘work’ here?** We cannot tell from one interval; that is what the Monte Carlo study measures.

5.6 Rubin’s rules in a picture (from Step 5.8f)

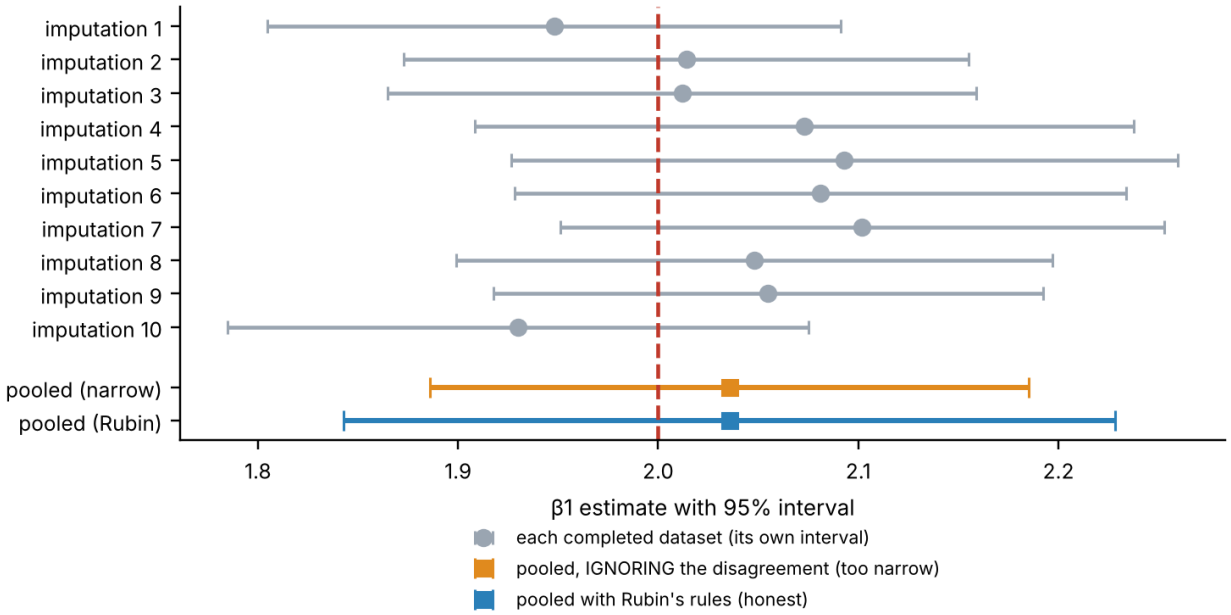


Figure 5.6: The 10 completed datasets of MICE (normal) on the MAR walkthrough dataset, and the pooled interval.

How to read it	Ten grey rows: each completed dataset’s estimate and interval. Orange: the pooled interval if the disagreement between them were ignored. Blue: the Rubin interval.
What we see	The ten estimates range from 1.930 to 2.102; pooled 2.036; SE without the disagreement 0.076, with Rubin’s rules 0.098 (29% wider).
Why	Different plausible fillings give different answers; that disagreement is real uncertainty. Single imputation has one filling, so it cannot see it.
Say it	“MICE tries several plausible versions of the missing values and lets their disagreement widen the interval.”

5.7 EM convergence

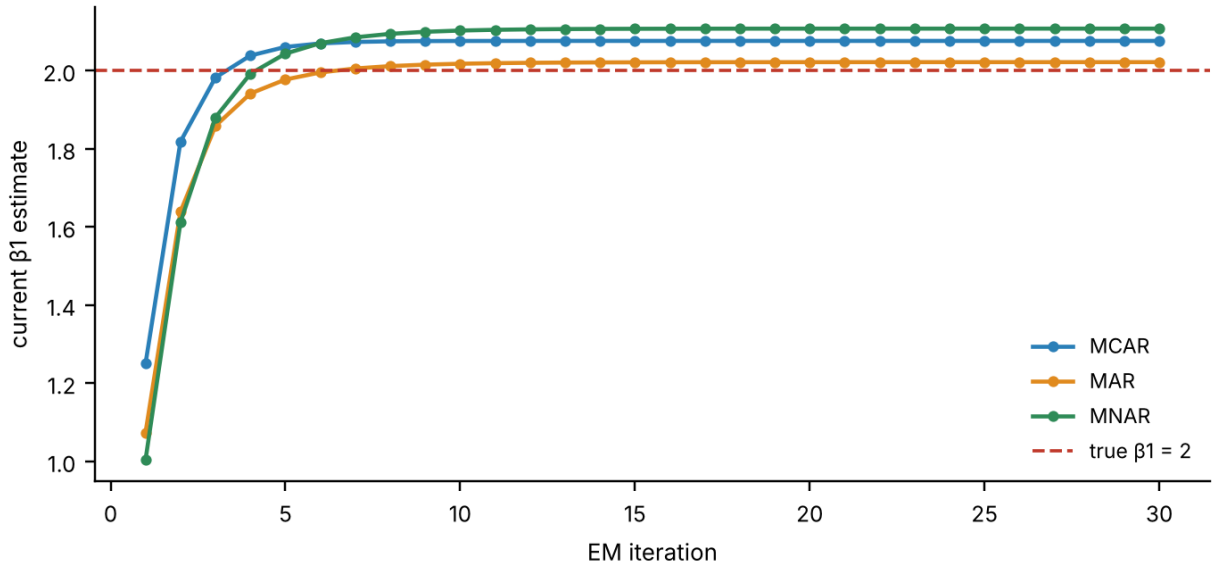


Figure 5.7: The EM estimate of β1 after each iteration, for the three walkthrough datasets.

How to read it	Horizontal: iteration; vertical: current estimate. A flat line means EM has converged.
What we see	Each line settles quickly; the change falls below 0.001 after 9–13 iterations. Final values equal the EM estimates in Table 4.7 (2.076, 2.022, 2.108).
Say it	“EM is a loop that stops when the answer stops changing.”
If asked	**Is the converged answer always right?** It is the maximum-likelihood answer if EM’s assumptions (normality, MAR) hold; under MNAR it converges to a biased number.

5.8 The Monte Carlo results, one mechanism (Steps 8g, 8j, 8m; report Figures 4–6)

The notebook draws three separate figures per mechanism; the report combines them into one three-panel figure. MAR is explained in full; MCAR and MNAR have the same layout.

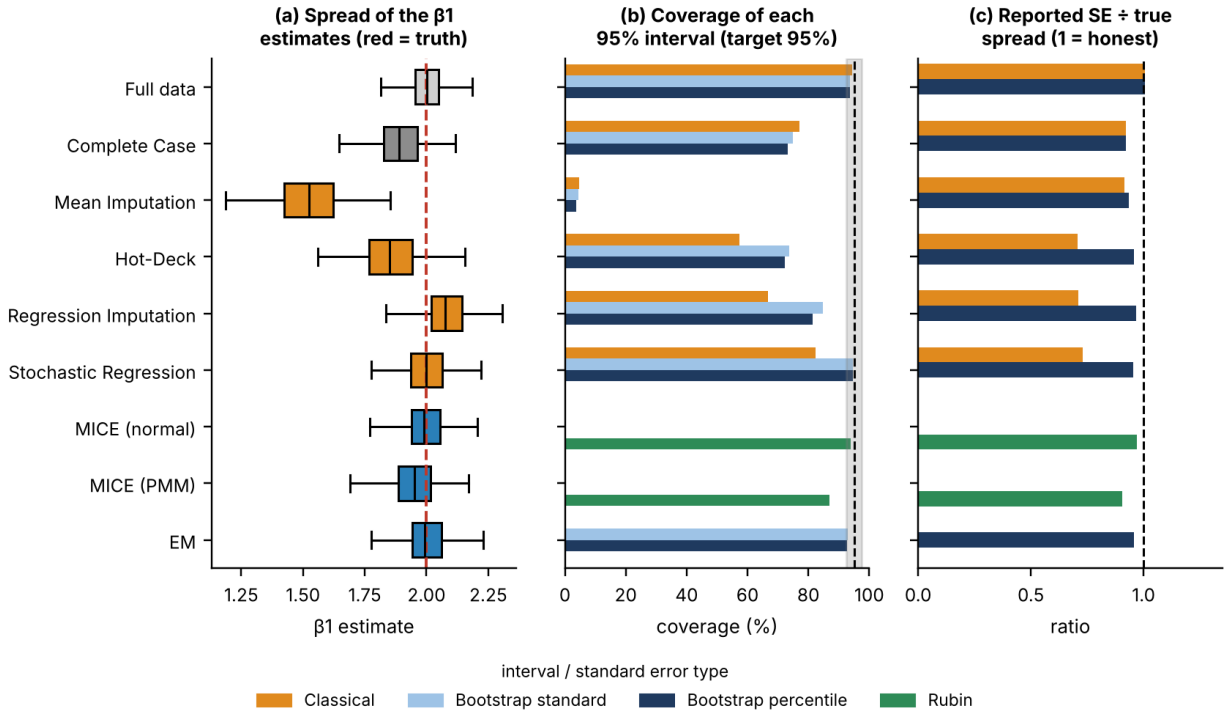


Figure 5.8: MAR. (a) spread of the 300 estimates; (b) coverage of each interval type; (c) reported SE ÷ true spread.

Panel (a)	One **box plot** per method of the 300 estimates: the box holds the middle half, the line inside is the median, whiskers the typical range. **Centred on the red line = unbiased; narrow = precise.**
Panel (b)	For each method, one bar per interval it has (orange classical, light blue bootstrap standard, dark blue bootstrap percentile, green Rubin). Dashed line = 95%; grey band = random error (±2.5). **A bar in the band = honest interval.**
Panel (c)	Reported SE ÷ true spread for the classical (orange), bootstrap (dark blue) and Rubin (green) standard errors. **1 = honest; below 1 = over-confident.**
What we see	(a) Complete case sits below 2 (bias -0.103); mean imputation far below; EM, MICE normal and stochastic regression centred on 2. (b) Stochastic regression: classical 82% → bootstrap 95%; complete case stays around 73–77% with any interval; EM 93%, MICE normal 94%, MICE PMM 87%. (c) Classical SE below 0.85 for Hot-Deck, Regression Imputation, Stochastic Regression; bootstrap SE between 92% and 99% of the spread for every method.
Why	Single imputation treats guesses as data, so its classical SE is too small; the bootstrap repeats the imputation and fixes the SE. Complete case's SE is fine, but its estimate is biased because the rows that lose values are not a random subset — no interval can fix that.

Say it	“Read left to right: is the box on the line, is the bar at 95, is the ratio near 1? Under MAR only EM, MICE normal and stochastic regression with the bootstrap pass all three.”
If asked	**Why does complete case’s coverage not improve with the bootstrap?** Its SE was already honest; its problem is bias, and the bootstrap resamples the same biased data.

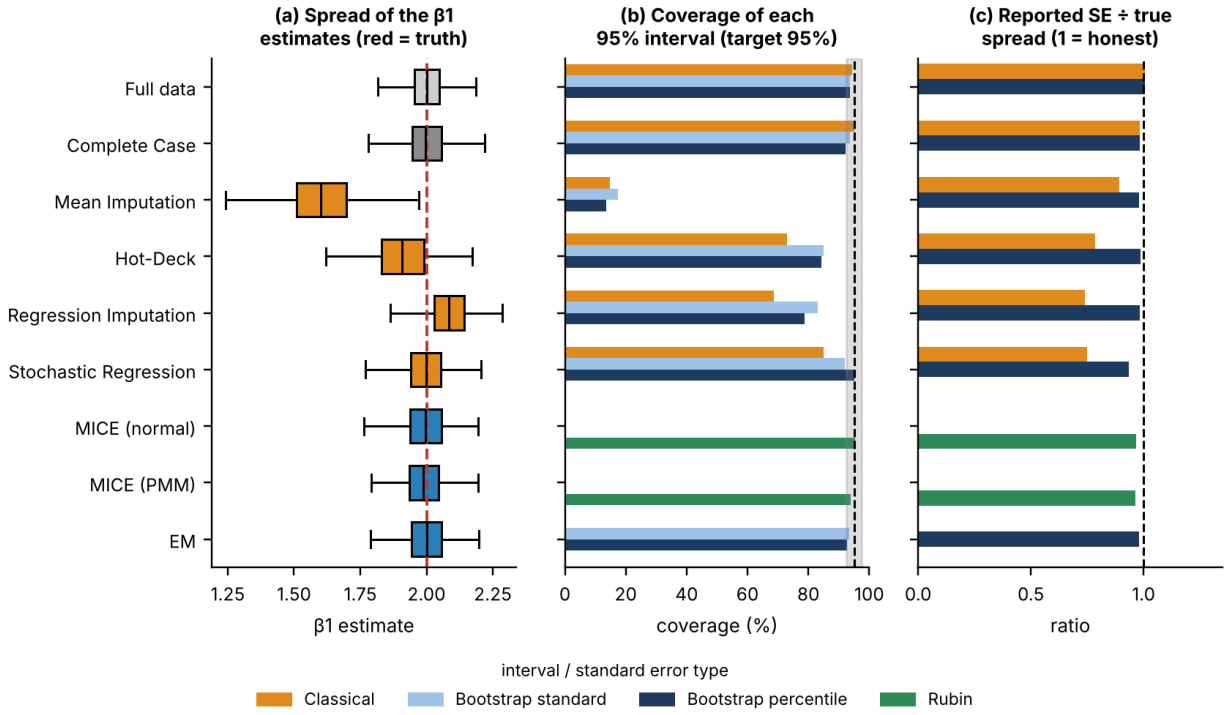


Figure 5.9: MCAR (same layout).

What to look for	Complete case, MICE and EM healthy: boxes on the line, bars near 95 (95%, 95%, 93%). Stochastic regression jumps from 85% to 95% with the bootstrap; hot-deck and regression improve but remain biased.
Say it	“Even under MCAR, single imputation’s classical interval is too narrow; the bootstrap repairs it when the method is unbiased.”

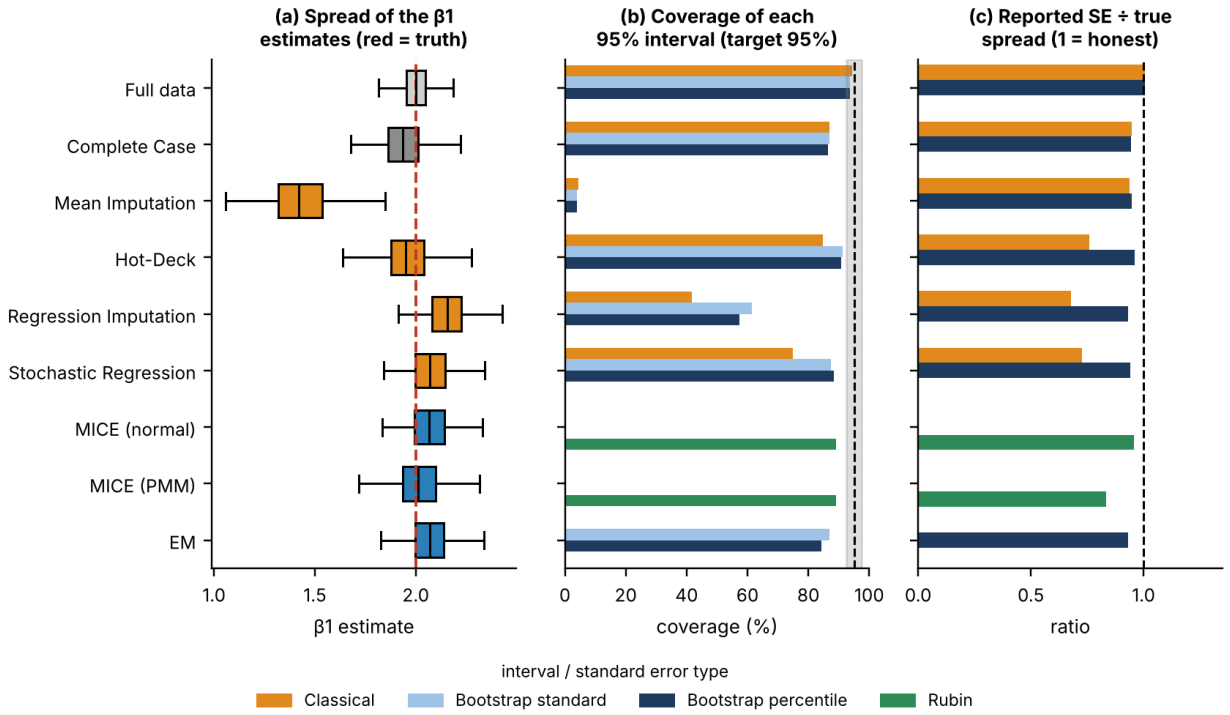


Figure 5.10: MNAR (same layout).

What to look for	No bar reaches the grey band. Complete case's box is left of the line, EM/MICE/stochastic boxes to the right — errors in opposite directions. Highest coverage of any interval: 91%.
Say it	"When missingness depends on the missing value, every method is off — and the bootstrap cannot fix it."

5.9 Individual intervals: classical vs bootstrap (Step 8n; report Figure 7)

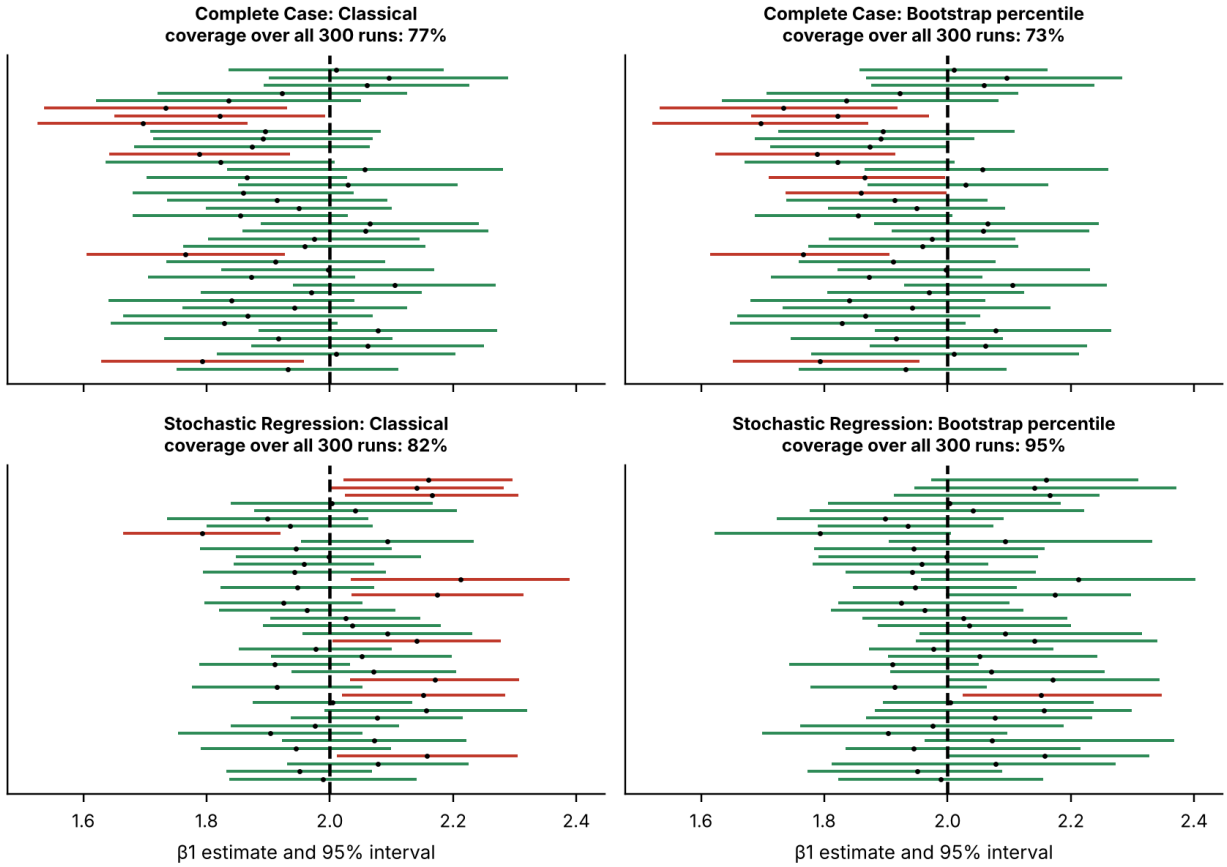


Figure 5.11: MAR, the first 40 intervals of two methods. Left classical, right bootstrap percentile. Green contains 2, red misses.

How to read it	Each horizontal line is the interval from ONE simulated dataset; the dot is its estimate; the dashed line is the truth. The title gives coverage over all 300.
What we see	Complete case misses mostly on the low side, classical (6 below, 0 above) and bootstrap (8 below, 0 above). Stochastic regression's classical intervals miss on both sides (1 below, 9 above); its bootstrap intervals are wider and miss less (0 below, 1 above).
Why	Bias moves intervals to one side; an over-confident SE makes them too short. The bootstrap fixes only the second.
Say it	"A 95% interval should miss about 2 times in 40. Complete case keeps missing low whatever we do; stochastic regression stops missing once the bootstrap widens it."

5.10 Before and after the bootstrap (Step 8o; report Figure 8)

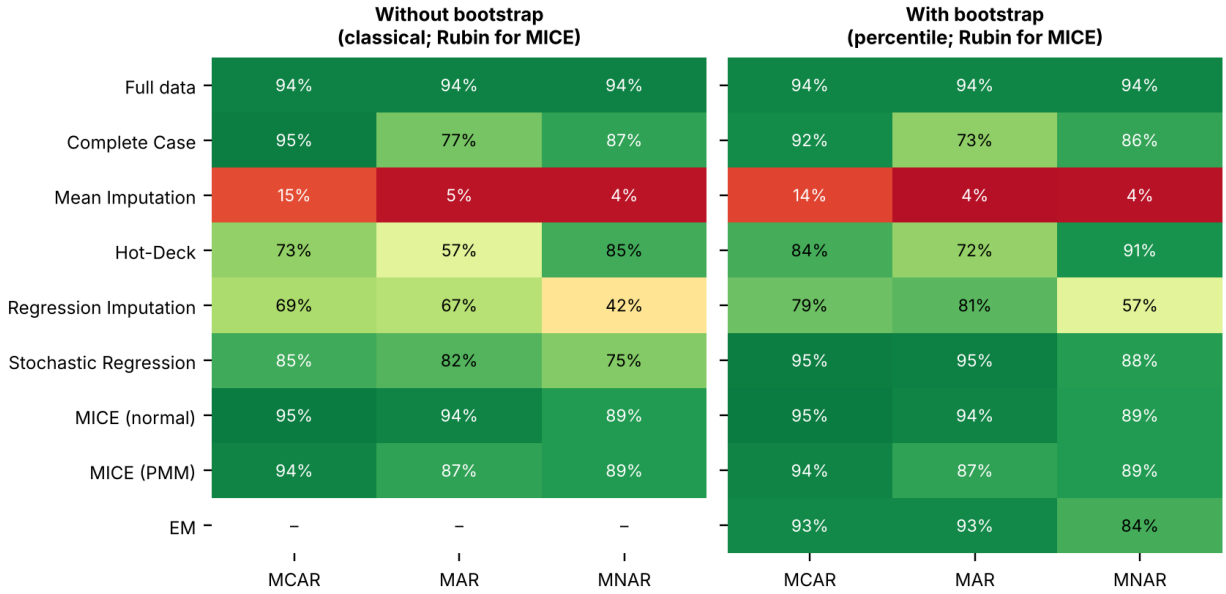


Figure 5.12: Coverage heat maps without (left) and with (right) the bootstrap. MICE uses the Rubin interval in both.

How to read it	Rows are methods, columns mechanisms, each cell a coverage %. Dark green ≈ 95%, through yellow to red as coverage falls; "–" = interval not available (EM has no classical interval). **Read down a column to compare methods within a mechanism; compare left and right to see what the bootstrap changes.**
What we see	Stochastic regression turns dark green under MCAR and MAR on the right; hot-deck and regression get lighter but stay below 95; mean imputation stays red; complete case under MAR stays light (77% → 73%); the MNAR column has no dark-green cell.
Say it	"Left to right is the effect of the bootstrap: it repairs over-confidence, never bias."

5.11 What the bootstrap changes, method by method (explainer only)

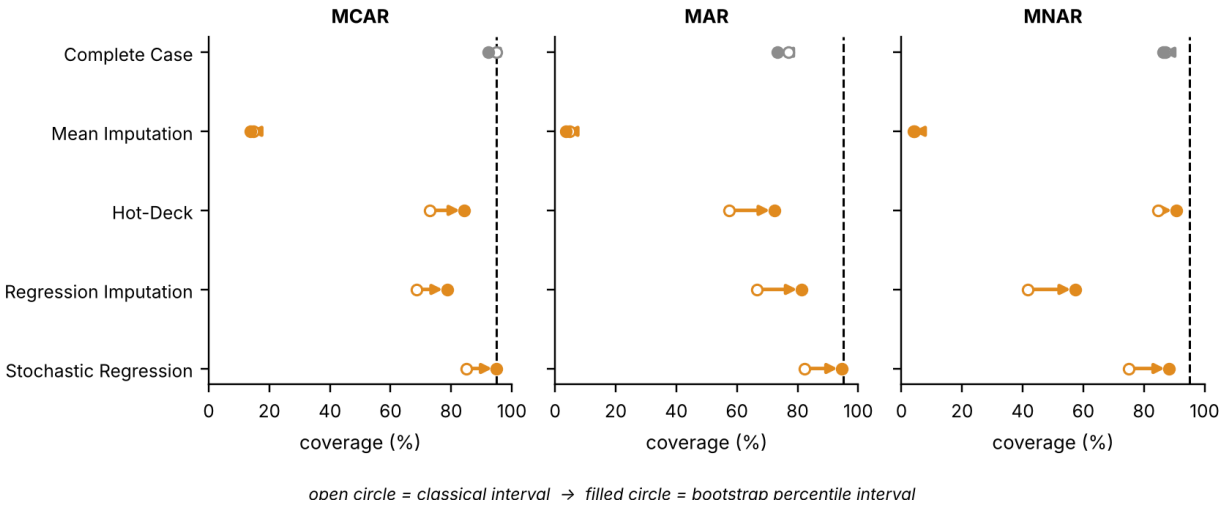


Figure 5.13: Coverage of the classical interval (open circle) and the bootstrap percentile interval (filled circle) for the five methods that have both.

How to read it	Each arrow goes from the classical to the bootstrap coverage of one method; the dashed line is 95%. A long arrow ending near 95 = the bootstrap fixed it.
What we see	Under MCAR and MAR stochastic regression's arrow reaches 95; hot-deck and regression move right but stop short; mean imputation and complete case barely move.
Say it	"The arrow is long only where the method was unbiased but over-confident."

5.12 Why bias limits coverage (explainer only; report Table 16)

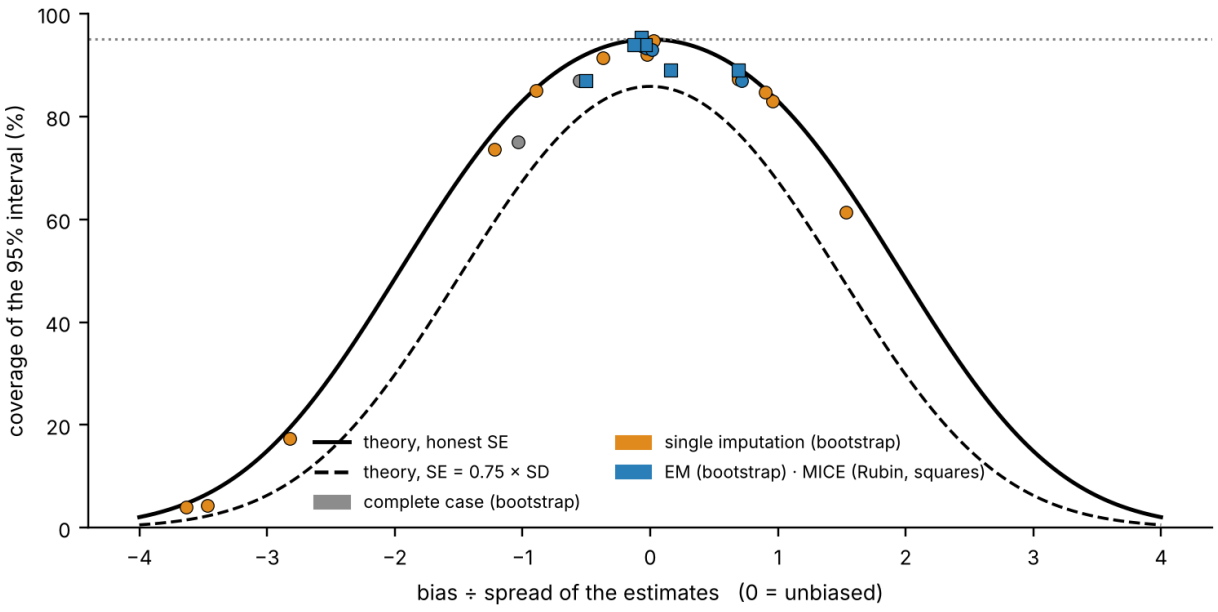


Figure 5.14: Coverage against bias ÷ spread. Lines: theory. Dots: the Monte Carlo results.

How to read it	Horizontal: how far the average estimate is from the truth, in units of its spread. Vertical: coverage. Solid line: theory with an honest SE; dashed: theory with an SE 25% too small. Dots: each method under each mechanism (bootstrap standard interval; squares = MICE with Rubin).
What we see	The dots lie on the solid line: once the SE is honest, coverage is decided by bias alone. Predicted and observed coverage agree within 2.8 points for the methods checked in Table 6.4.
Say it	“This one curve explains the project: honest SE plus zero bias gives 95%; any bias pulls coverage down the curve.”
If asked	**Where does the formula come from?** If the estimate is normal with bias b and SD, the interval covers 2 exactly when |estimate − 2| ≤ 1.96·SE, which gives $\Phi(1.96 \cdot r - b/SD) - \Phi(-1.96 \cdot r - b/SD)$.

5.13 Where each figure is explained

Notebook step / report	Figure	Here
2e	the data	5.1
report Figure 1	deletion rules	5.2
4i	missing counts	5.3
5.10d–f / report Figure 2	what each method did	5.4
6d / report Figure 3	one bootstrap	5.5
(from 5.8f)	Rubin’s rules	5.6
(explainer)	EM convergence	5.7
8g, 8j, 8m / report Figures 4–6	box plots, coverage, SE honesty	5.8
8n / report Figure 7	individual intervals	5.9
8o / report Figure 8	heat maps before/after	5.10
(explainer)	classical → bootstrap arrows; bias vs coverage	5.11, 5.12

Table 5.1: Figure map.

6 The Results in Plain Words

Results from 300 Monte Carlo datasets, one mechanism at a time. Remember: **methods are compared inside a mechanism**. Good values: bias near 0, coverage near 95 (within ±2.5), SE ÷ spread near 1.

6.1 MCAR: the gaps are pure luck

Method	Bias	RMSE	Classical	Boot. standard	Boot. percentile	Rubin
Full data (benchmark)	-0.000	0.071	94.3	93.7	93.7	–
Complete Case	-0.005	0.091	95.0	93.7	92.3	–
Mean Imputation	-0.397	0.421	14.7	17.3	13.7	–
Hot-Deck	-0.096	0.144	73.0	85.0	84.3	–
Regression Imputation	+0.081	0.117	68.7	83.0	78.7	–
Stochastic Regression	-0.002	0.095	85.0	92.0	95.0	–
MICE (normal)	-0.006	0.087	–	–	–	95.3
MICE (PMM)	-0.011	0.087	–	–	–	94.0
EM	-0.003	0.084	–	93.3	92.7	–

Table 6.1: MCAR. Bias, RMSE and coverage (%) of every interval.

The story. Rows that remain are a fair sample, so **complete case is fine** (bias -0.005, coverage 95%); its cost is precision. **MICE and EM** are unbiased, honest and slightly more accurate. **Stochastic regression** is unbiased, but its classical interval covers only 85%; with the bootstrap, 95%. Hot-deck and regression imputation are biased, so the bootstrap helps but cannot reach 95%. Mean imputation fails completely.

Say it like this. “Under MCAR, deleting incomplete rows is safe. Filling gaps once is not — unless you use a bootstrap that repeats the filling-in.”

6.2 MAR: the gaps depend on something we observe

Method	Bias	RMSE	Classical	Boot. standard	Boot. percentile	Rubin
Full data (benchmark)	-0.000	0.071	94.3	93.7	93.7	–
Complete Case	-0.103	0.144	77.0	75.0	73.3	–
Mean Imputation	-0.485	0.505	4.7	4.3	3.7	–
Hot-Deck	-0.151	0.196	57.3	73.7	72.3	–
Regression Imputation	+0.082	0.123	66.7	84.7	81.3	–
Stochastic Regression	+0.003	0.098	82.3	94.7	94.7	–
MICE (normal)	-0.003	0.090	–	–	–	94.0
MICE (PMM)	-0.050	0.112	–	–	–	87.0
EM	+0.001	0.090	–	93.0	92.7	–

Table 6.2: MAR. Bias, RMSE and coverage (%) of every interval.

The story. Rows with large Z — and therefore large Y — lose values, so the complete rows are not a fair sample: **complete case is biased** (-0.103) and every interval covers below 80%. Its SE is honest; the problem is where the interval is centred. **EM and MICE (normal)** use Z, are essentially unbiased, most accurate, and cover 93% and 94%. **Stochastic regression with the bootstrap** also works (82% → 95%). MICE (PMM) is slightly biased (87%).

Say it like this. “When missingness depends on a variable we can see, we must use it. EM and MICE do; complete case cannot — and no interval fixes that.”

6.3 MNAR: the gaps depend on the missing value itself

Method	Bias	RMSE	Classical	Boot. standard	Boot. percentile	Rubin
Full data (benchmark)	-0.000	0.071	94.3	93.7	93.7	–
Complete Case	-0.060	0.124	87.0	87.0	86.3	–
Mean Imputation	-0.573	0.594	4.3	4.0	4.0	–
Hot-Deck	-0.047	0.135	84.7	91.3	90.7	–
Regression Imputation	+0.157	0.188	41.7	61.3	57.3	–
Stochastic Regression	+0.074	0.129	75.0	87.3	88.3	–
MICE (normal)	+0.068	0.120	–	–	–	89.0
MICE (PMM)	+0.019	0.119	–	–	–	89.0
EM	+0.071	0.122	–	87.0	84.3	–

Table 6.3: MNAR. Bias, RMSE and coverage (%) of every interval.

The story. Nothing reaches 95%; the highest coverage of any interval is 91%. Complete case is biased down, the MAR-based methods (EM, MICE normal, stochastic) up. The bootstrap helps a little but cannot remove MNAR bias. Hot-deck’s small bias here is most likely errors cancelling.

Say it like this. “When the reason for missingness is the missing value, the observed data cannot tell us how to correct. Report a sensitivity analysis.”

6.4 The one table that explains it all

Mechanism	Method	Bias ÷ SD	Boot. SE ÷ SD	Predicted coverage	Observed coverage
MCAR	Hot-Deck	-0.90	0.99	84.7%	85.0%
MCAR	Mean Imputation	-2.82	0.98	18.4%	17.3%
MAR	Complete Case	-1.03	0.92	77.8%	75.0%
MAR	Hot-Deck	-1.22	0.96	74.5%	73.7%
MAR	Regression Imputation	+0.90	0.97	83.7%	84.7%
MAR	Stochastic Regression	+0.03	0.95	93.8%	94.7%
MNAR	Regression Imputation	+1.53	0.93	61.3%	61.3%
MNAR	Stochastic Regression	+0.69	0.94	87.0%	87.3%

Table 6.4: Coverage of the bootstrap-standard interval, predicted from bias and SD alone (Section 3.14), and observed.

Once the bootstrap makes the SE honest, coverage is almost exactly what the bias predicts. **The bootstrap repairs variance, not bias.**

6.5 What holds in every mechanism

- Mean imputation is worst (no interval covers more than 17.3%).
- Classical intervals after single imputation are too narrow (SE 68–79% of the spread).
- Bootstrap SEs are honest for every method (92–99% of the spread).
- Rubin’s rules (MICE) and EM give honest intervals under MCAR and MAR.
- With nothing missing, classical and bootstrap intervals cover 94.3% and 93.7% — the simulation and the bootstrap themselves work.

6.6 Why the one-dataset walkthrough is not evidence

On the walkthrough dataset complete case under MNAR gives 2.015, close to 2 — tempting to call it fine. Over 300 datasets it is biased (-0.060). **The walkthrough is for understanding; the Monte Carlo is the evidence.**

7 How to Present This Project

7.1 A five-minute talk

Time	Say	Show
0:00–0:30	Question: does the way we handle missing data change a regression estimate, and can we trust its interval?	Figure 1.1
0:30–1:15	Simulated data, true β1 = 2, n = 200, helper Z; 20% of Y and X1 deleted under MCAR, MAR, MNAR.	Figure 5.2
1:15–2:00	Eight methods in three families: delete rows; fill once; fill many times / likelihood.	Figure 5.4
2:00–2:45	Intervals: classical, bootstrap (standard, percentile), Rubin. Bootstrap = one dataset → one interval; Monte Carlo = 300 datasets → coverage. We nested them: N × B.	Figure 3.2
2:45–4:15	Results per mechanism; the bootstrap repairs over-confidence but not bias.	Figures 5.8 and 5.12
4:15–5:00	Conclusions and limitations.	Table 7.2

Table 7.1: A five-minute talk.

7.2 The summary table to quote

Mechanism	What works	What to avoid
MCAR	Complete case (classical interval); MICE or EM for a little more precision.	Mean imputation; classical intervals after single imputation.
MAR	EM (bootstrap) or MICE normal (Rubin), with the helper variable; stochastic regression only with a bootstrap interval.	Complete case (biased); single imputation with classical intervals.
MNAR	Nothing is reliable; report a sensitivity analysis.	Claiming any method or interval fixes MNAR.

Table 7.2: Recommendations by mechanism.

7.3 Explaining the design to your supervisor

These are the exact points from your discussion with sir, and how the project now answers each one:

Sir’s point	How the project handles it
Bootstrap and Monte Carlo are different things.	Both are used and kept separate: the bootstrap makes one interval per dataset; Monte Carlo draws new datasets from the true model (Section 3.7).
Without Monte Carlo you cannot find the coverage probability.	Coverage is measured only from the 300 Monte Carlo datasets; the one-dataset walkthrough makes no coverage claims.
A single bootstrap gives one interval and tells nothing more.	Notebook Section 6 says exactly this and then moves to the Monte Carlo.
Bootstrap intervals need N × B replications.	N = 300 datasets × B = 200 resamples for six methods (≈ 1 million fits).
There are four bootstrap intervals; only standard and t use the bootstrap SE.	We use standard and percentile; bootstrap-t and BCa are listed as limitations.
Different intervals give different coverage.	Coverage is reported separately for classical, bootstrap standard, bootstrap percentile and Rubin.
Parallel computing reduces the burden.	The simulation runs in parallel with joblib (Python’s equivalent of R’s parallel tools), and saves progress so it can resume.

Table 7.3: The supervisor’s comments and the answers in the project.

7.4 Do’s and don’ts

Do	Don’t
Say “within each mechanism”.	Rank the mechanisms against each other.
Say “the bootstrap repairs the SE, not bias”.	Say the bootstrap “fixes missing data”.
Say coverage comes from Monte Carlo.	Claim one bootstrap interval shows a method works.
Say “cannot be separated” when within random error.	Declare a winner from a 2–3 point difference.
Mention the strong helper variable Z.	Claim the results hold for every real dataset.

Table 7.4: Do’s and don’ts.

8 Likely Questions and Model Answers

About the design

Q. Why simulated data?

With real data the true answer is unknown, so we cannot say which method is right. With simulated data we chose β1 = 2 and can measure exactly how close each method gets and how often its interval contains it.

Q. What is the difference between the bootstrap and Monte Carlo?

The bootstrap resamples the one dataset we have to produce one standard error and one interval; it does not know the truth. Monte Carlo draws many new datasets from the true model; because the truth is known, it can count how often intervals contain it.

Q. Why put the bootstrap inside the Monte Carlo?

To find the coverage of bootstrap intervals we need many bootstrap intervals, one per dataset, each checked against the truth. So for each of 300 datasets we bootstrap 200 times: N × B.

Q. Why N = 300 and B = 200?

300 datasets make coverage accurate to about ±2.5 points; 200 resamples give a stable bootstrap SE. Together the run takes about 40 minutes on one core, less in parallel. Larger values would be smoother but slower.

Q. Why delete values in two variables?

If only Y is missing, complete case, regression imputation and EM give identical coefficients and hide their differences. With X1 also missing, the methods genuinely differ.

Q. What is Z?

A noisy copy of Y, observed for everyone, not in the regression. It stands for auxiliary information. Ours is strong (correlation 0.94); real ones are usually weaker.

Q. Why compare within mechanisms?

The mechanism is a property of how data went missing, not a choice. The practical question is: given my situation, which method should I use?

About intervals and coverage

Q. What is coverage?

The share of the 300 intervals that contain the true value. A 95% interval should cover about 95%.

Q. What is the classical interval and why is it ‘naive’ after imputation?

Estimate ± 1.96 × the regression-formula SE. After single imputation the formula treats made-up values as real, so it thinks it has more information than it does and the SE is too small.

Q. What are the bootstrap standard and percentile intervals?

Standard: estimate ± 1.96 × bootstrap SE (the SD of the B bootstrap estimates). Percentile: the 2.5% and 97.5% points of the B bootstrap estimates.

Q. Why did you not use bootstrap-t or BCa?

Bootstrap-t needs a second bootstrap inside every resample (N × B × B′ fits), far too slow here; BCa needs extra jackknife computations. We list both as limitations.

Q. What is the Rubin interval?

For MICE: pooled estimate ± 1.96 × √(W + (1 + 1/m)·B), where W is the average within-dataset variance and B the variance between the m estimates.

Q. Did the bootstrap work?

It made the SE honest for every method (92–99% of the true spread), so unbiased methods reach about 95%. It could not fix biased methods.

Q. Why can the bootstrap not fix bias?

It resamples the observed data; a method that is biased on the data is equally biased on every resample, so the interval is centred in the wrong place.

Q. How do you know bias explains the remaining shortfall?

The formula $\Phi(1.96 \cdot r - b/SD) - \Phi(-1.96 \cdot r - b/SD)$ predicts the coverage from bias and spread alone; it matches the observed coverage within 2.8 points (Table 6.4).

Q. With nothing missing, coverage is below 95. Is something wrong?

No: 94.3% (classical) and 93.7% (bootstrap) are within the random error of ±2.5 points. Bootstrap intervals are also known to be slightly narrow in moderate samples.

About the methods

Q. Why is mean imputation biased?

All filled values equal the average, so the filled rows carry no relationship between Y and X1; the slope is pulled towards zero.

Q. Why is complete case biased under MAR?

Rows with large Z lose values, and Z is linked to Y even given the predictors, so the rows kept are not a fair sample for the regression.

Q. Why is regression imputation biased upward?

Filled values sit exactly on the line, with no scatter, which makes the relationship look stronger.

Q. Stochastic regression is unbiased — why does its classical interval under-cover?

Its SE is too small (73% of the spread under MAR) because it fills each gap once and treats it as real. The bootstrap fixes this: coverage rises to 95%.

Q. Single imputation + bootstrap or MICE?

Both work when unbiased. MICE needs 10 imputations; the bootstrap needs 200 complete re-runs. Rubin's rules are the cheaper route; the bootstrap is the general one (it works for EM too).

Q. Why is PMM worse than normal MICE under MAR?

The data are exactly normal, so the normal model is right; PMM can only borrow observed values, and the rows MAR targets have few similar donors.

Q. What does EM do?

It finds the means and covariances that make the observed data most likely, alternating an expectation step (fill gaps with expected values) and a maximization step (re-estimate), then computes the regression coefficients.

About the conclusions

Q. Why does every method fail under MNAR?

The reason for missingness is the missing value itself, which the observed data cannot reveal; nothing in the data tells us how much to correct.

Q. What should be done under MNAR?

A sensitivity analysis: repeat the analysis under several plausible assumptions about the missing values and report how the conclusions change.

Q. How do you know the simulation is correct?

Built-in checks: our regression matches statsmodels; EM with nothing missing equals OLS; the deletion rules were verified on 20,000 rows; with nothing missing the intervals cover 94.3% and 93.7%; the resumed run gave identical results.

Q. What are the main limitations?

One simple model; chosen mechanism strengths; a very strong helper variable; B = 200 makes percentile limits a little noisy; no bootstrap-t or BCa; simplified MICE and EM; only $\beta 1$ examined; Monte Carlo error.

Q. Which method do you recommend?

MCAR: complete case, or MICE/EM. MAR: EM or MICE normal with auxiliary variables (or stochastic regression with a bootstrap interval). MNAR: none — sensitivity analysis. Never mean imputation; never classical intervals after single imputation.

Appendix A: Glossary

Term	Meaning
Auxiliary variable	A variable related to the incomplete one, not in the analysis model, used to help fill gaps (Z).
Bias	Average of (estimate − true value).
Bootstrap	Resampling rows of one dataset with replacement and redoing the analysis, to get an SE and interval from that dataset.
Bootstrap standard interval	Estimate ± 1.96 × bootstrap SE.
Bootstrap percentile interval	2.5% and 97.5% points of the bootstrap estimates.
Bootstrap-t, BCa	Refined bootstrap intervals (not used).
Classical interval	Estimate ± 1.96 × formula SE.
Complete case	Keep only rows with no missing values.
Coverage	Share of repeated intervals that contain the true value.
EM	Expectation–maximization: maximum likelihood with incomplete data.
Hot-deck	Copy the value of the most similar row.
MAR / MCAR / MNAR	Missing at random / completely at random / not at random.
MICE	Multiple imputation by chained equations.
Monte Carlo	Drawing many datasets from a known model to study how a method behaves.
Monte Carlo error	Random error of a simulation result from a finite number of repetitions.
N × B replication	N Monte Carlo datasets, each bootstrapped B times.
Naive SE	Formula SE that ignores that values were filled in.
Over-confident	Reported SE smaller than the true spread.
Parallel computing	Running independent repetitions on several CPU cores at once (joblib).
PMM	Predictive mean matching: borrow a real value from a row with a close prediction.
RMSE	Typical size of the error; combines bias and spread.
Rubin’s rules	Pooling for multiple imputation: total = W + (1 + 1/m)·B.
Seed	Starting number of a random-number generator.
Sensitivity analysis	Repeating an analysis under different assumptions about the missing data.
Single imputation	Fill each gap once and treat it as real.
Spread (SD)	SD of the estimates across repetitions.
Standard error	Typical wobble of an estimate between datasets.
Stochastic regression	Regression prediction plus a random residual.

Appendix B: Formula Sheet

Idea	Formula
Model	$Y = 5 + 2 \cdot X1 - 1.5 \cdot X2 + X3 + \varepsilon$; $Z = Y + \text{noise}$
Bias, RMSE	$\text{bias} = \text{mean}(\beta 1 - 2)$; $\text{RMSE} = \sqrt{\text{mean}((\beta 1 - 2)^2)} \approx \sqrt{\text{bias}^2 + \text{SD}^2}$
Classical / bootstrap-standard / Rubin interval	estimate ± 1.96 × SE
Bootstrap SE	SD of the B bootstrap estimates
Percentile interval	2.5% and 97.5% points of the bootstrap estimates
Rubin's rules	β = mean of m estimates; $T = W + (1 + 1/m) \cdot B$; $SE = \sqrt{T}$
Coverage	100 × (number of intervals containing 2) / N
Monte Carlo error	$\sqrt{p(1 - p)/N} \approx 1.3$ points for $p = 0.95$, $N = 300$
Deletion chance (MAR, MNAR)	$\text{expit}(-2.23 + 2 \cdot d)$, $\text{expit}(z) = 1/(1 + e^{(-z)})$
Coverage with bias	$\Phi(1.96 \cdot r - b/SD) - \Phi(-1.96 \cdot r - b/SD)$, $r = SE/SD$
EM → coefficients	$\text{slopes} = \Sigma xx^{-1} \Sigma xy$; $\text{intercept} = \text{mean}(Y) - \text{slopes} \cdot \text{mean}(X)$

Appendix C: Notebook Map

Steps	What they do	Explained in
1a–1b	tools and settings	4.1
2a–2e	complete data	3.1, 4.2, 5.1
3a–3c	regression helper, checks, baseline	4.3
4a–4j	deletion rules, three datasets, checks	2.4, 3.12, 4.4, 5.3
5.1–5.9	the eight methods on one dataset	3.8–3.11, 4.5
5.10a–f	what the methods did	4.5, 5.4
6a–6d	bootstrap and intervals on one dataset	3.6, 4.6, 5.5
7a–7g	the Monte Carlo study (N × B)	3.7, 4.7
8a–8o	results per mechanism, intervals, heat maps	5.8–5.10, 6
9a–9g	master tables, findings	4.8, 6

Appendix D: The Repository

github.com/swachchhogun/missing-data-stat-inference

Folder / file	Contents
report/	the project report (PDF)
notebook/	the notebook with all outputs
results/	monte_carlo_results.csv (every repetition), settings, summary tables
figures/	report figures
report_build/	scripts that rebuild figures, report and README from the notebook and results
requirements.txt	packages needed

- **Run:** pip install -r requirements.txt, open the notebook, Run all. With USE_SAVED_RESULTS = True the saved simulation is reloaded instantly; set it to False (or change a setting) to recompute (about 40 minutes on one core, faster with more cores).
- **Quick test:** N_REPS = 20, B_BOOT = 50.
- **If interrupted:** run again — it resumes from the saved progress.